

Analysis of the Obstacle Potential Function

1 Potential Function

$$U_{obst} = \frac{\lambda(|\Delta v|/\Delta v_{max})}{k(\Delta x, \Delta y, \Delta v_x, \Delta v_y)} \quad (1)$$

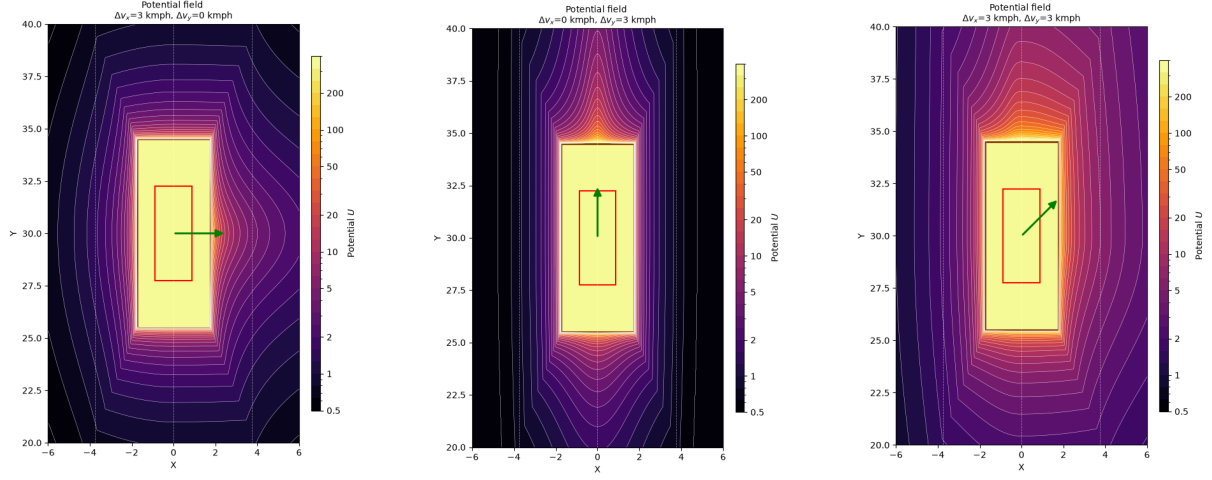
$$k(\Delta x, \Delta y, \Delta v_x, \Delta v_y) = \begin{cases} 0, & \text{if } |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{if } |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases} \quad (2)$$

$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (3)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha|\Delta v_y|) + (\alpha|\Delta v_y| - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (4)$$

Here $\Delta x, \Delta y$ are the relative position wrt obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities wrt obstacle vehicle.

1.1 Potential Field Plots



(a) Varying Δv_x

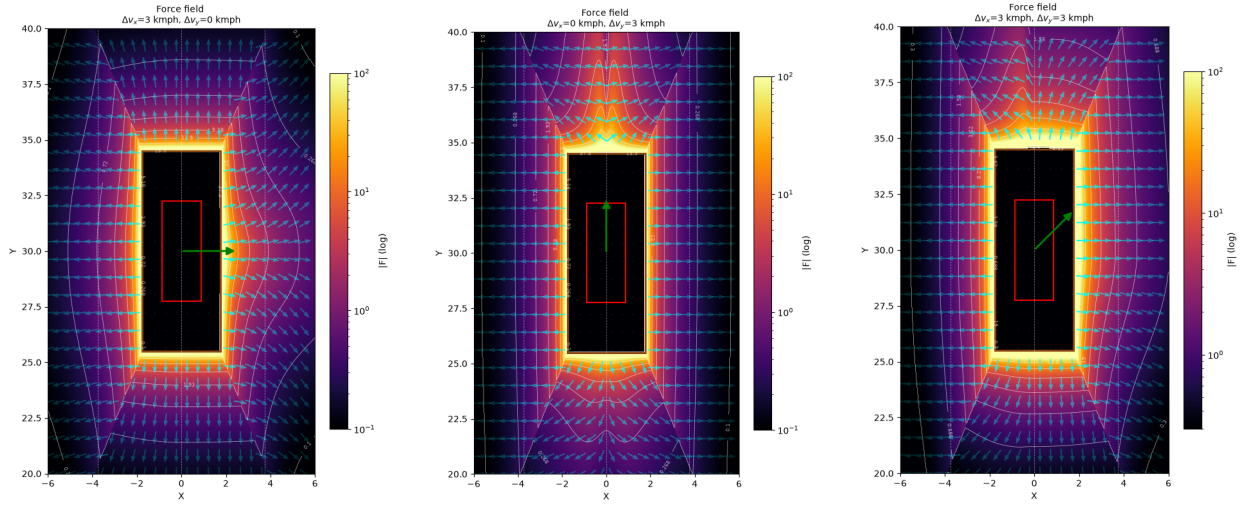
(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 1: Obstacle potential field U_{obst} .

1.2 Force Field

$$F = -\nabla U_{obst}$$



(a) Varying Δv_x

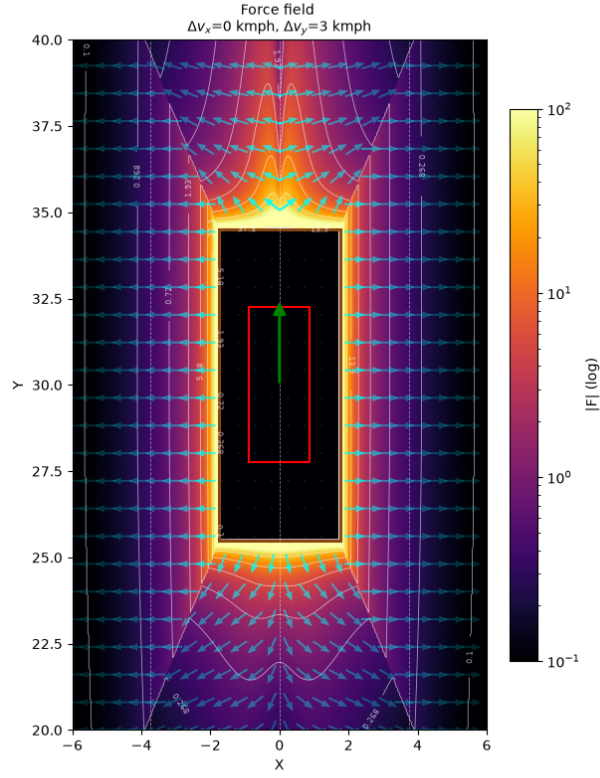
(b) Varying Δv_y

(c) Varying $\Delta v_x, \Delta v_y$

Figure 2: Force field derived from U_{obst} .

2 Possible Issues

2.1 Anomalous dip in the magnitude of force field



The magnitude of the force field dips near $x=0$ in the above plot which is undesirable as the magnitude of force field should be maximum along $x=0$ when compared to the neighbouring points.

2.2 Sensitivity to τ_x, τ_y and Parallel Forces

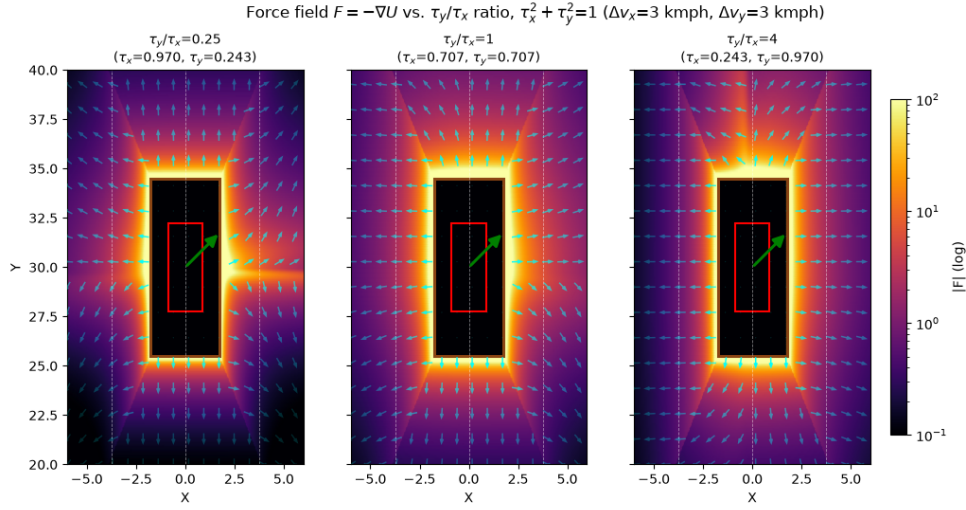


Figure 3: Sensitivity of the field to τ_x, τ_y .

The shape of the potential is highly sensitive to τ_y/τ_x and very high value or a very small value can cause the force field to be parallel in certain region which is not desirable.

2.3 Velocity Behavior at a fixed point

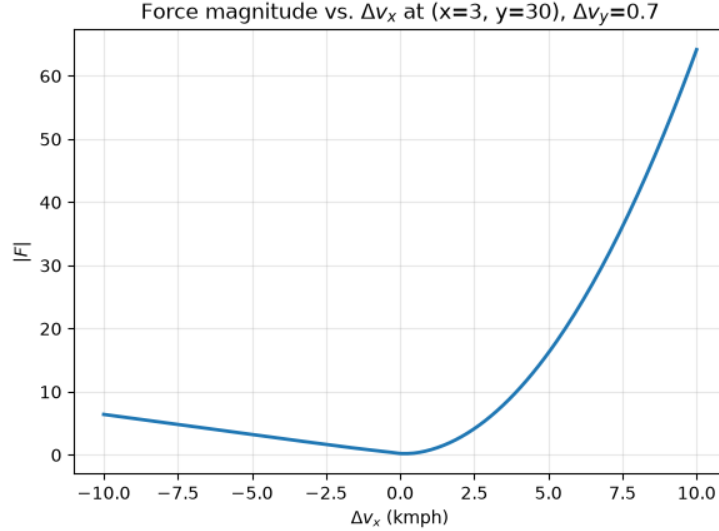


Figure 4: Velocity at a fixed point.

The force increases with increase in receding velocity at a fixed point. In an ideal scenario the repulsive force should decrease when the vehicles are receding.

2.4 More free parameters than DOF

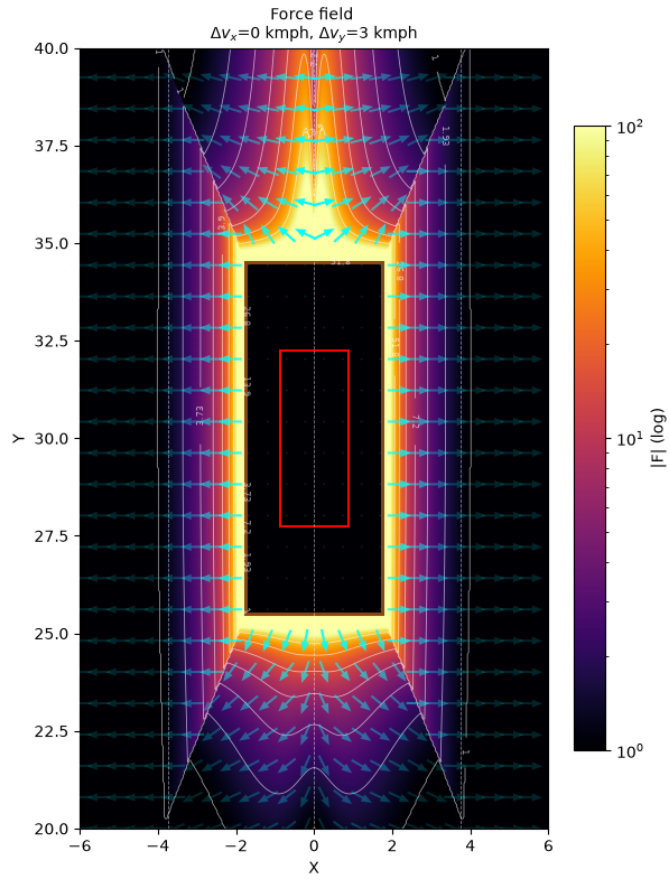
The model introduces 4 free parameters ($\lambda, \alpha, \tau_x, \tau_y$) while the degree of freedom of the model is only 3. So reduction of parameters will help in the optimization problem.

3 Potential Fix

3.1 Anomalous dip in the magnitude of force field

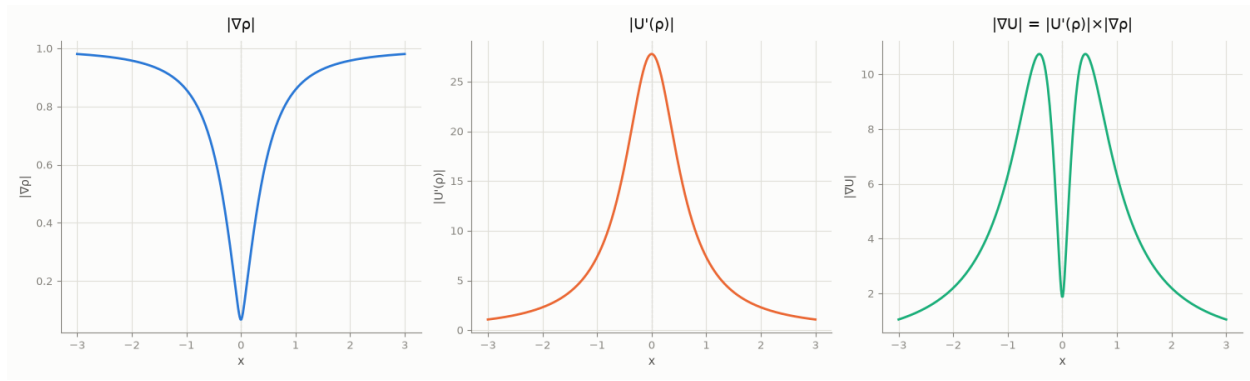
3.1.1 The Cause

$$U(k) = \frac{\lambda (|\Delta v|/\Delta v_{max})}{k}, \quad F = -\nabla U.$$



$$\nabla U = U'(\rho) \nabla k$$

$$|\nabla U| = |U'(k)| \cdot |\nabla k|,$$

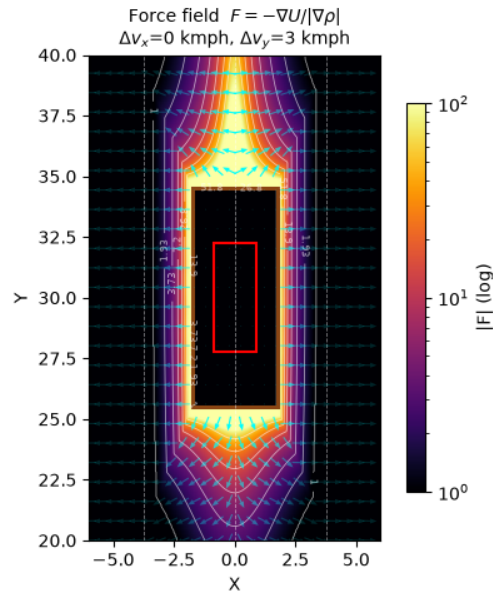


3.1.2 The fix

$$U'(k) = -\frac{\lambda(|\Delta v|/\Delta v_{max})}{k^2}.$$

$U'(k)$ falls off smoothly with k and has no dip or ridge of its own.

$$F = -\frac{\nabla U}{|\nabla \rho|} = -U'(\rho) \frac{\nabla \rho}{|\nabla \rho|}, \quad |F| = |U'(\rho)|.$$



3.2 Sensitivity to τ_x, τ_y

$$U = \frac{\lambda(\Delta v/v)}{k}$$

$$k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\frac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}, & |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau_x}\right)^2 + \left(\frac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases}$$

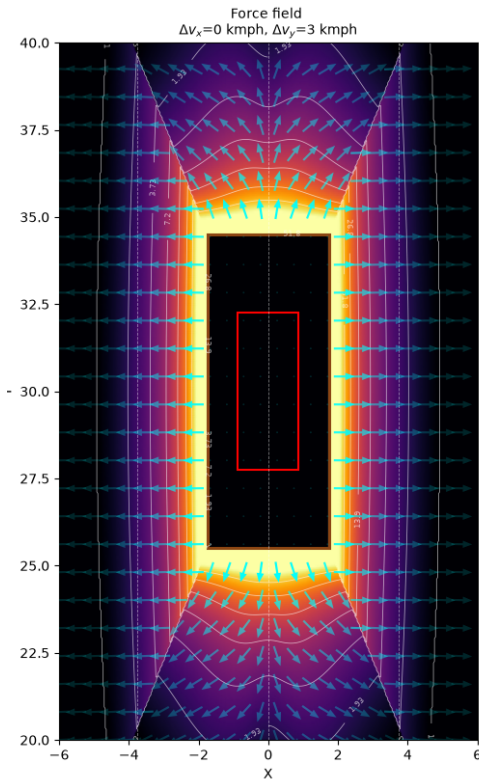
If $\tau_y/\tau_x \ll 1$, i.e. $\tau_y \ll \tau_x$, the y^2/τ_y^2 term dominates. The potential then does not change across x , so the lateral force is absent in the forward and backward regions. The same issue occurs in the side region when $\tau_y/\tau_x \gg 1$.

3.2.1 The Fix

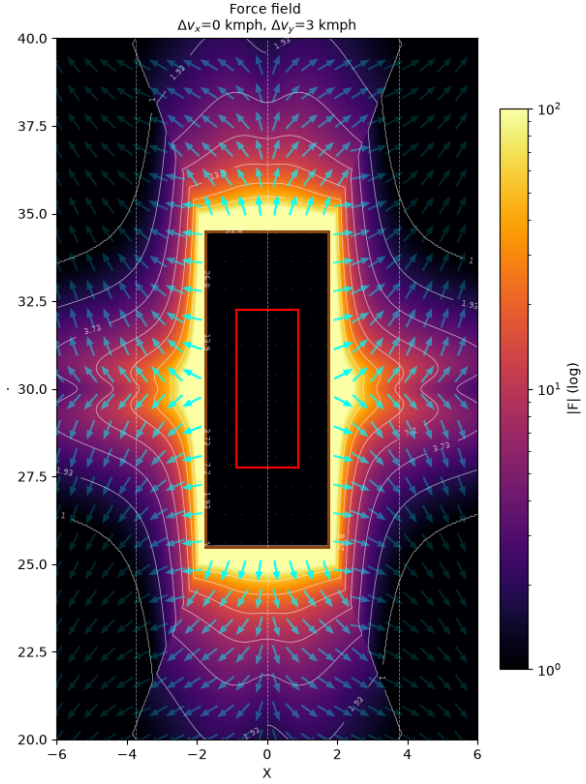
k splits the space into different regions, so different τ values can be used for different regions:

$$K = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{\left(\frac{\Delta x}{\tau(\Delta x, \Delta v_x, \tau_2)}\right)^2 + \left(\frac{\Delta y}{\tau(\Delta y, \Delta v_y, \tau_1)}\right)^2}, & |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{\left(\frac{\Delta x}{\tau(\Delta x, \Delta v_x, \tau_1)}\right)^2 + \left(\frac{\Delta y}{\tau(\Delta y, \Delta v_y, \tau_2)}\right)^2}, & \text{otherwise} \end{cases}$$

$$\tau(\Delta x, \Delta v_x, \tau) = \frac{\tau}{2} \left[(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1) \tanh(-\Delta x \Delta v_x) \right]$$



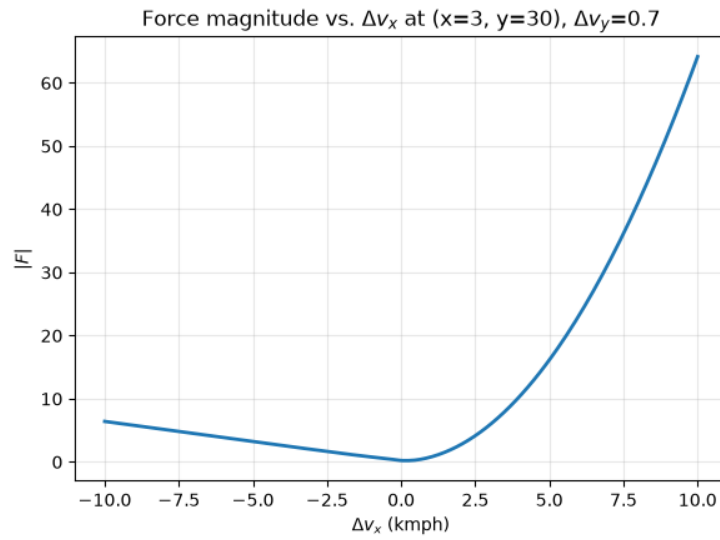
(a) $\tau_y/\tau_x > 5$



(b) $\tau_2/\tau_1 > 5$

3.3 Velocity Behaviour

3.3.1 The Issue

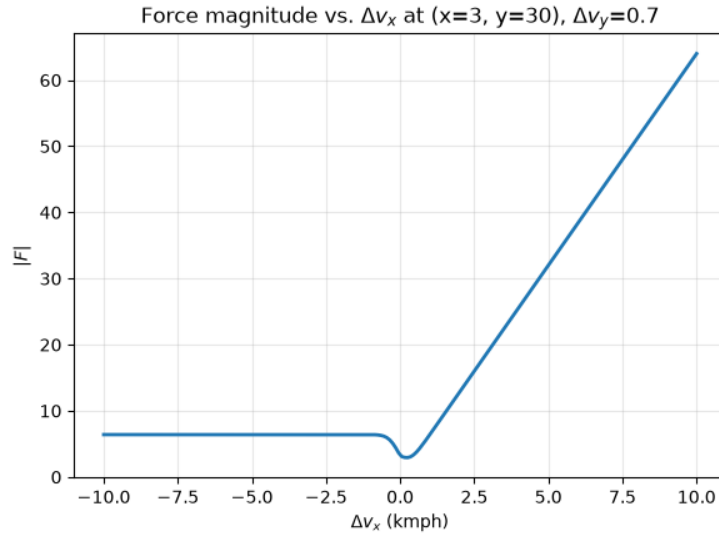


Velocity increases with increase in receding velocity is due to the $|\Delta v|/\Delta v_{max}$ term in the numerator of the potential function.

3.3.2 The Fix

Remove that term from numerator

$$U(k) = \frac{\lambda}{k},$$



The kink in the above plot is due to the term

$$\tau = \left((1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1) \tanh(-\Delta x \Delta v_x) \right) \tau_x$$

The original goal was to have

$$\tau = \tau_x \text{ (vehicle receding),} \quad \tau = \alpha \tau_x |\Delta v_x| \text{ (vehicle approaching)}$$

Its better to have τ greater in cases when vehicles are approaching.

$$\tau = \tau_x \text{ (vehicle receding),} \quad \tau = \tau_x (1 + \alpha|\Delta v_x|) \text{ (vehicle approaching)}$$

Formulation 1:

$$\tau = \left((1 + (1 + \alpha|\Delta v_x|)) + ((1 + \alpha|\Delta v_x|) - 1) \tanh(-\Delta x \Delta v_x) \right) \tau_x$$

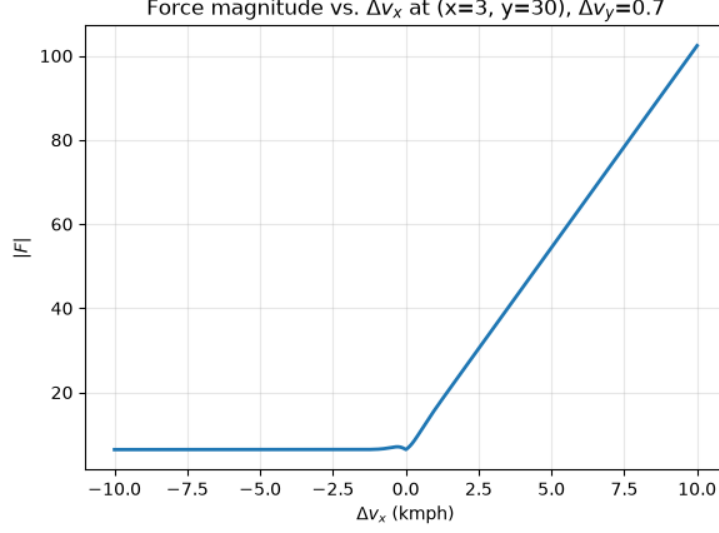


Figure 6: Force magnitude vs. Δv_x under Formulation 1.

We want the repulsive force to decrease with increase in receding velocity.

Formulation 2:

$$\tau = \left(\left(\frac{1}{1 + \alpha|\Delta v_x|} + (1 + \alpha|\Delta v_x|) \right) + \left((1 + \alpha|\Delta v_x|) - \frac{1}{1 + \alpha|\Delta v_x|} \right) \tanh(-\Delta x \Delta v_x) \right) \tau_x$$

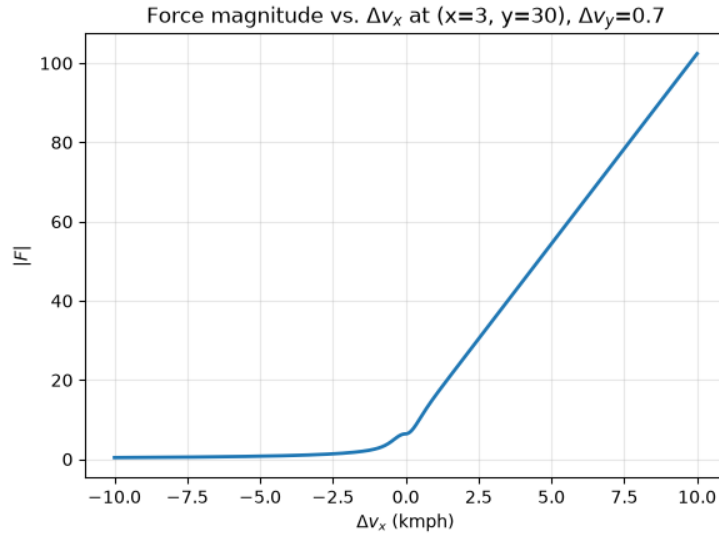


Figure 7: Force magnitude vs. Δv_x under Formulation 2.

Now if consider the case for high approaching velocity, we have $\tau \approx \alpha|\Delta v_x| \tau_x$. Here the params α and τ_x are kind of redundant. We could otherwise use α to control the power to which the approaching velocity affects τ .

Formulation 3:

$$\tau = \left(\left(\frac{1}{(1 + |\Delta v_x|)^\alpha} + (1 + |\Delta v_x|)^\alpha \right) + \left((1 + |\Delta v_x|)^\alpha - \frac{1}{(1 + |\Delta v_x|)^\alpha} \right) \tanh(-\Delta x \Delta v_x) \right) \tau_x$$

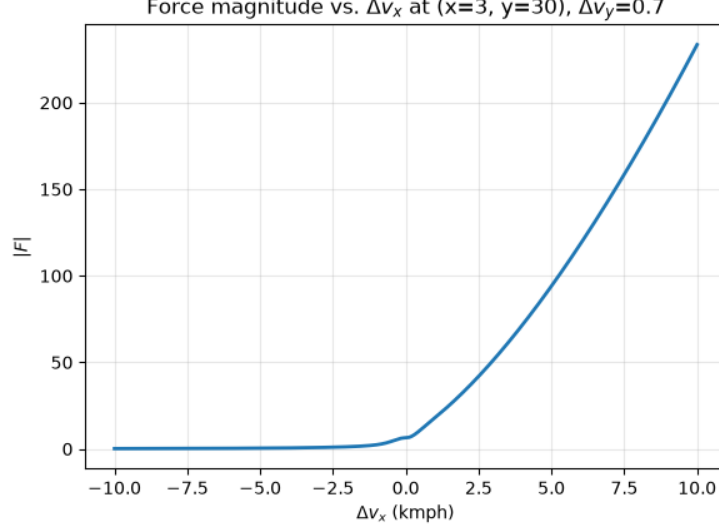
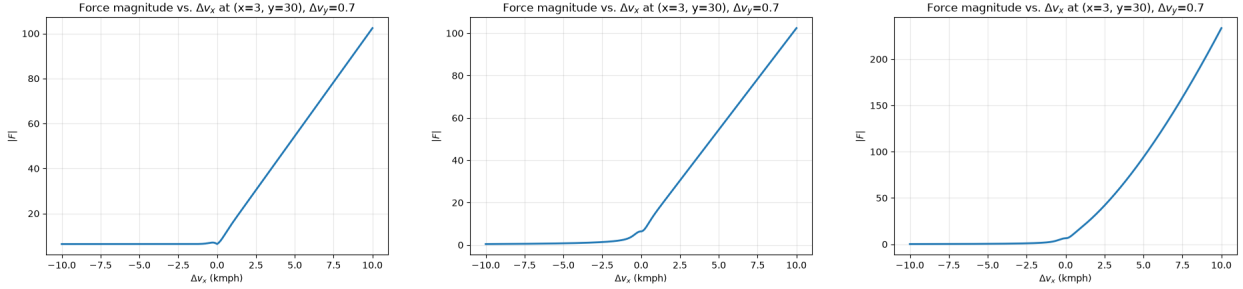


Figure 8: Force magnitude vs. Δv_x under Formulation 3.



(a) Formulation 1

(b) Formulation 2

(c) Formulation 3

Figure 9: Comparison of force magnitude vs. Δv_x across the three formulations.

3.4 Free Parameters

$$\gamma = \frac{\tau_y}{\tau_x}, \quad \beta = \lambda \tau_x$$

$$U = \frac{\beta(\Delta v/v)}{k}$$

$$k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \frac{w}{|\Delta x|}\right) \sqrt{(\Delta x)^2 + \left(\frac{\Delta y}{\tau(\gamma)}\right)^2}, & |\Delta y| \leq \frac{l|\Delta x|}{w} \\ \left(1 - \frac{l}{|\Delta y|}\right) \sqrt{(\Delta x)^2 + \left(\frac{\Delta y}{\tau(\gamma)}\right)^2}, & \text{otherwise} \end{cases}$$

This reduces the free parameters from λ, τ_x, τ_y to γ and β .